

# Estimation Theory

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2024 2nd Semester

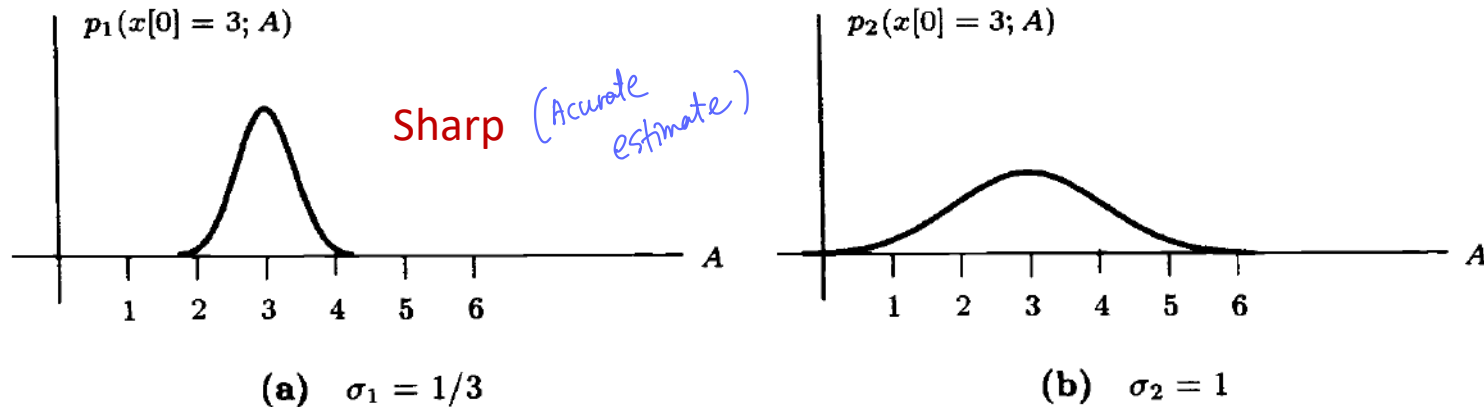
# **CRAMER-RAO LOWER BOUND (CRLB)**

# Estimator Accuracy

$$x[0] = A + w[0], \quad w[0] \sim \mathcal{N}(0, \sigma^2)$$

Likelihood – distribution of data given parameters

$$p(x[0]; A) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x[0] - A)^2\right)$$



Large curvature -> more precise estimation of  $A$  is possible

# Curvature of the Log-Likelihood Function

Likelihood:  $p(x[0]; A) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x[0] - A)^2\right)$

Log-likelihood:  $\ln p(x[0]; A) = -\ln \sqrt{2\pi\sigma^2} - \frac{1}{2\sigma^2}(x[0] - A)^2$   
*(constant)*

$$\frac{\partial \ln p(x[0]; A)}{\partial A} = \frac{1}{\sigma^2}(x[0] - A)$$

Curvature:  $-\frac{\partial^2 \ln p(x[0]; A)}{\partial A^2} = \frac{1}{\sigma^2}$

As  $\sigma^2 \downarrow 0$ , the curvature  $\uparrow$  (better estimation).

For  $\hat{A} = x[0]$ ,  $\text{var}(\hat{A}) = \sigma^2$ .  $\longrightarrow$   $\text{var}(\hat{A}) = \frac{1}{-\frac{\partial^2 \ln p(x[0]; A)}{\partial A^2}}$  *Inverse of curvature*

As  $\text{var}(\hat{A}) \downarrow 0$ , the curvature  $\uparrow$  (better estimation).

Measure of curvature:  $-\mathbb{E}\left(\frac{\partial^2 \ln p(x[0]; A)}{\partial A^2}\right)$  ; Average curvature of log-likelihood

# Theorem: Cramer-Rao Lower Bound (CRLB)

**Theorem:** It is assumed that the probability density function  $p(x; \theta)$  satisfies the *regularity* condition

$$\mathbb{E} \left( \frac{\partial \ln p(x; \theta)}{\partial \theta} \right) = 0 \quad \text{for all } \theta, \quad (1)$$

where the expectation is taken with respect to  $p(x; \theta)$ . Then, the variance of any unbiased estimator  $\hat{\theta}$  must satisfy

$$\text{var}(\hat{\theta}) \geq \frac{1}{-\mathbb{E} \left( \frac{\partial^2 \ln p(x; \theta)}{\partial \theta^2} \right)}. \quad (2)$$

# Additional Facts about the CRLB

1. An unbiased estimator achieves the bound if and only if

$$\frac{\partial \ln p(x; \theta)}{\partial \theta} = I(\theta)(g(x) - \theta)$$

for some functions  $I$  and  $g$ .

2.  $\hat{\theta} = g(x)$  is the minimum variance unbiased estimator.

3. The minimum variance is  $1/I(\theta)$ , where

$$I(\theta) = \mathbb{E} \left( \left( \frac{\partial \ln p(x; \theta)}{\partial \theta} \right)^2 \right) = -\mathbb{E} \left( \frac{\partial^2 \ln p(x; \theta)}{\partial \theta^2} \right).$$

↑  
Fisher Information

*← curvature*

# Regularity Condition

$$\mathbb{E} \left( \frac{\partial \ln p(x; \theta)}{\partial \theta} \right) = 0 \quad \text{for all } \theta$$

$$\begin{aligned} \mathbb{E} \left( \frac{\partial \ln p(x; \theta)}{\partial \theta} \right) &= \int \frac{\partial \ln p(x; \theta)}{\partial \theta} p(x; \theta) dx \\ &= \int \frac{1}{p(x; \theta)} \frac{\partial p(x; \theta)}{\partial \theta} p(x; \theta) dx \\ &= \int \frac{\partial p(x; \theta)}{\partial \theta} dx \\ &= \frac{\partial}{\partial \theta} \int p(x; \theta) dx \\ &= 0 \end{aligned}$$

$$\frac{\partial}{\partial \theta} \left( \int \frac{\partial \ln p(x; \theta)}{\partial \theta} p(x; \theta) dx = 0 \right)$$

$$\int \left( \frac{\partial^2 \ln p(x; \theta)}{\partial \theta^2} p(x; \theta) + \frac{\partial \ln p(x; \theta)}{\partial \theta} \frac{\partial p(x; \theta)}{\partial \theta} \right) dx = 0$$

$$\mathbb{E} \left( \frac{\partial^2 \ln p(x; \theta)}{\partial \theta^2} \right) + \int \frac{\partial \ln p(x; \theta)}{\partial \theta} \frac{\partial \ln p(x; \theta)}{\partial \theta} p(x; \theta) dx = 0$$

Hence, 
$$\mathbb{E} \left( \left( \frac{\partial \ln p(x; \theta)}{\partial \theta} \right)^2 \right) = -\mathbb{E} \left( \frac{\partial^2 \ln p(x; \theta)}{\partial \theta^2} \right).$$

# Proof of CRLB

Let  $\alpha = g(\theta)$ . Consider all unbiased estimators  $\hat{\alpha}$  such that

$$\mathbb{E}(\hat{\alpha}) = \alpha = g(\theta).$$

$$\frac{\partial}{\partial \theta} \left( \int \hat{\alpha} p(x; \theta) dx = g(\theta) \right) \quad \int \hat{\alpha} \frac{\partial p(x; \theta)}{\partial \theta} dx = \frac{\partial g(\theta)}{\partial \theta}$$

$$\int \hat{\alpha} \frac{\partial \ln p(x; \theta)}{\partial \theta} p(x; \theta) dx = \frac{\partial g(\theta)}{\partial \theta}$$


$$\alpha \mathbb{E} \left( \frac{\partial \ln p(x; \theta)}{\partial \theta} \right) = 0 \quad \text{regularity condition}$$

$$\int (\hat{\alpha} - \alpha) \frac{\partial \ln p(x; \theta)}{\partial \theta} p(x; \theta) dx = \frac{\partial g(\theta)}{\partial \theta}$$



# Cauchy-Schwarz Inequality

For vectors:

$$|\langle x, y \rangle|^2 \leq \|x\|^2 \|y\|^2$$

$$|\langle x, y \rangle| = \|x\| \|y\| |\cos \theta| \leq \|x\| \|y\|$$

equality if  $x = cy$ .

*x and y are parallel*

For random variables (r.v.):

$\mathbb{E}(XY) = \langle X, Y \rangle$  ; an inner product between two mean zero r.v.'s

$|\mathbb{E}(XY)|^2 \leq \mathbb{E}(X^2)\mathbb{E}(Y^2)$  ; Cauchy-Schwarz Inequality

$$\left| \int X(z)Y(z)p(z)dz \right|^2 \leq \left( \int X^2(z)p(z)dz \right) \cdot \left( \int Y^2(z)p(z)dz \right)$$

# Back to the Proof of CRLB

We have

$$\int (\hat{\alpha} - \alpha) \frac{\partial \ln p(x; \theta)}{\partial \theta} p(x; \theta) dx = \frac{\partial g(\theta)}{\partial \theta}$$

Cauchy-Schwarz:  $\left| \int X(z)Y(z)p(z)dz \right|^2 \leq \left( \int X^2(z)p(z)dz \right) \cdot \left( \int Y^2(z)p(z)dz \right)$

Let  $X(z) \leftarrow \hat{\alpha} - \alpha$ ,  $Y(z) \leftarrow \frac{\partial \ln p(x; \theta)}{\partial \theta}$ , and  $p(z) \leftarrow p(x; \theta)$ .

$$\begin{aligned} \left( \frac{\partial g(\theta)}{\partial \theta} \right)^2 &= \left( \int (\hat{\alpha} - \alpha) \frac{\partial \ln p(x; \theta)}{\partial \theta} p(x; \theta) dx \right)^2 \\ &\leq \underbrace{\left( \int (\hat{\alpha} - \alpha)^2 p(x; \theta) dx \right)}_{\text{var}(\hat{\alpha})} \left( \int \left( \frac{\partial \ln p(x; \theta)}{\partial \theta} \right)^2 p(x; \theta) dx \right) \end{aligned}$$

Hence,

$$\text{var}(\hat{\alpha}) \geq \frac{\left( \frac{\partial g(\theta)}{\partial \theta} \right)^2}{\mathbb{E} \left( \left( \frac{\partial \ln p(x; \theta)}{\partial \theta} \right)^2 \right)} = \frac{\left( \frac{\partial g(\theta)}{\partial \theta} \right)^2}{-\mathbb{E} \left( \frac{\partial^2 \ln p(x; \theta)}{\partial \theta^2} \right)}$$

Let  $\alpha = g(\theta) = \theta$ ,  
we get the desired result.

# Proof of Additional Results

The Cauchy-Schwarz inequality becomes an equality if  $X(z) = cY(z)$  for some constant  $c$ , not depending on  $z$ . (parallel)

In our case, this is when 
$$\frac{\partial \ln p(x; \theta)}{\partial \theta} = \frac{1}{c}(\hat{\alpha} - \alpha)$$

$$\frac{\partial \ln p(x; \theta)}{\partial \theta} = \frac{1}{c(\theta)}(\hat{\theta} - \theta) \quad \text{for } \alpha = g(\theta) = \theta$$

To find  $c(\theta)$ :

$$\frac{\partial^2 \ln p(x; \theta)}{\partial \theta^2} = \frac{\partial}{\partial \theta} \left( \frac{1}{c(\theta)} \right) (\hat{\theta} - \theta) - \frac{1}{c(\theta)}$$

$$-\mathbb{E} \left( \frac{\partial^2 \ln p(x; \theta)}{\partial \theta^2} \right) = \frac{1}{c(\theta)}$$

Hence,

$$c(\theta) = \frac{1}{-\mathbb{E} \left( \frac{\partial^2 \ln p(x; \theta)}{\partial \theta^2} \right)} = \frac{1}{I(\theta)} \quad \text{: fisher information}$$

$$\longrightarrow I(\theta) = \mathbb{E} \left( \left( \frac{\partial \ln p(x; \theta)}{\partial \theta} \right)^2 \right) = -\mathbb{E} \left( \frac{\partial^2 \ln p(x; \theta)}{\partial \theta^2} \right).$$

# Example 1

$$x[n] = A + w[n] \quad w[n] \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$$

$$\begin{aligned} \text{Estimator: } \tilde{A} &= \frac{1}{N} \sum_{n=0}^{N-1} x[n] & \mathbb{E}(\tilde{A}) &= A & \text{unbiased} \\ & & \text{var}(\tilde{A}) &= \frac{1}{N} \sigma^2 \end{aligned}$$

Likelihood of  $x := (x[0], \dots, x[N-1])$

$$\begin{aligned} p(x; A) &= \prod_{n=0}^{N-1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x[n] - A)^2\right) \\ &= \frac{1}{(2\pi\sigma^2)^{N/2}} \exp\left(-\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2\right) \end{aligned}$$

$$\frac{\partial \ln p(x; A)}{\partial A} = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} (x[n] - A) = \frac{N}{\sigma^2} (\bar{x} - A) \quad \bar{x} = \frac{1}{N} \sum_{n=0}^{N-1} x[n]$$

$$\frac{\partial \ln p(x; A)}{\partial A} = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} (x[n] - A) = \frac{N}{\sigma^2} (\bar{x} - A)$$

Regularity condition:  $\mathbb{E} \left( \frac{\partial \ln p(x; A)}{\partial A} \right) = 0$  (checked)

Second derivative:  $\frac{\partial^2 \ln p(x; A)}{\partial A^2} = -\frac{N}{\sigma^2}$

**CRLB:**  $\text{var}(\hat{A}) \geq \frac{\sigma^2}{N}$  for any unbiased estimator  $\hat{A}$

- Since  $\text{var}(\tilde{A}) = \frac{\sigma^2}{N}$ ,  $\tilde{A}$  is the minimum variance unbiased estimator.
- If an unbiased estimator achieves the CRLB, it is said to be *efficient*.

## Example 2

- Consider  $x[n] = s[n; \theta] + w[n]$ , for  $n = 0, 1, \dots, N - 1$ .
- Likelihood of  $\mathbf{x} := (x[0], \dots, x[N - 1])$  is

$$p(\mathbf{x}; \theta) = \frac{1}{(2\pi\sigma^2)^{N/2}} \exp \left( -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - s[n; \theta])^2 \right)$$

$$\frac{\partial \ln p(\mathbf{x}; \theta)}{\partial \theta} = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} (x[n] - s[n; \theta]) \frac{\partial s[n; \theta]}{\partial \theta}$$

$$\frac{\partial^2 \ln p(\mathbf{x}; \theta)}{\partial \theta^2} = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} \left( (x[n] - s[n; \theta]) \frac{\partial^2 s[n; \theta]}{\partial \theta^2} - \left( \frac{\partial s[n; \theta]}{\partial \theta} \right)^2 \right)$$

$$\mathbb{E} \left( \frac{\partial^2 \ln p(\mathbf{x}; \theta)}{\partial \theta^2} \right) = -\frac{1}{\sigma^2} \sum_{n=0}^{N-1} \left( \frac{\partial s[n; \theta]}{\partial \theta} \right)^2$$

$$\text{var}(\hat{\theta}) \geq \frac{\sigma^2}{\sum_{n=0}^{N-1} \left( \frac{\partial s[n; \theta]}{\partial \theta} \right)^2}$$

Special case:  $s[n; \theta] = \theta$ .  
Then  $\text{CRLB} = \frac{\sigma^2}{N}$ .

# **Transformation of Parameters**

$$x[n] = A + w[n], \quad \text{where } w[n] \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$$

Estimation of  $A^2$  ( $\alpha = g(A) = A^2$ )

$$\mathbf{var}(\hat{A}^2) \geq \frac{(2A)^2}{N/\sigma^2} = \frac{4A^2\sigma^2}{N}$$

CRLB for  $\alpha = g(\theta)$ :

$$\mathbf{var}(\hat{\alpha}) \geq \frac{\left(\frac{\partial g}{\partial \theta}\right)^2}{-\mathbb{E}\left(\frac{\partial^2 \ln p(x; \theta)}{\partial \theta^2}\right)}$$

### 1. Nonlinear transformation does not preserve efficiency

$$\bar{x} = \frac{1}{N} \sum_{n=0}^{N-1} x[n], \quad \bar{x} \sim \mathcal{N}\left(A, \frac{\sigma^2}{N}\right)$$

$$\text{Var}(X) = E(X^2) - \{E(X)\}^2$$

Is  $\bar{x}^2$  efficient for  $A^2$ ?

$$\begin{aligned} \mathbb{E}(\bar{x}^2) &= \mathbb{E}^2(\bar{x}) + \mathbf{var}(\bar{x}) = A^2 + \frac{\sigma^2}{N} \\ &\neq A^2. \end{aligned}$$

Biased => not efficient



$$x[n] = A + w[n], \quad \text{where } w[n] \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$$

Estimation of  $A^2$  ( $\alpha = g(A) = A^2$ )

$$\text{var}(\hat{A}^2) \geq \frac{(2A)^2}{N/\sigma^2} = \frac{4A^2\sigma^2}{N}$$

CRLB for  $\alpha = g(\theta)$ :

$$\text{var}(\hat{\alpha}) \geq \frac{\left(\frac{\partial g}{\partial \theta}\right)^2}{-\mathbb{E}\left(\frac{\partial^2 \ln p(x;\theta)}{\partial \theta^2}\right)}$$

## 2. Linear transform preserves efficiency

Assume:  $\hat{\theta}$  efficient for  $\theta$ ,  $g(\theta) = a\theta + b$  (affine transform)

$$\widehat{g(\theta)} = g(\hat{\theta}) = a\hat{\theta} + b$$

$$\begin{aligned} E(a\hat{\theta} + b) &= aE(\hat{\theta}) + b = a\theta + b \\ &= g(\theta) \end{aligned}$$

$$\text{var}(\widehat{g(\theta)}) = \text{var}(a\hat{\theta} + b) = a^2 \text{var}(\hat{\theta})$$

CRLB for  $g(\theta)$ :

$$\begin{aligned} \text{var}(\widehat{g(\theta)}) &\geq \frac{\left(\frac{\partial g}{\partial \theta}\right)^2}{I(\theta)} \\ &= \left(\frac{\partial g(\theta)}{\partial \theta}\right)^2 \text{var}(\hat{\theta}) \\ &= a^2 \text{var}(\hat{\theta}). \end{aligned}$$

Efficient

3. Efficiency is approximately maintained over a nonlinear transformation for a **large dataset**.

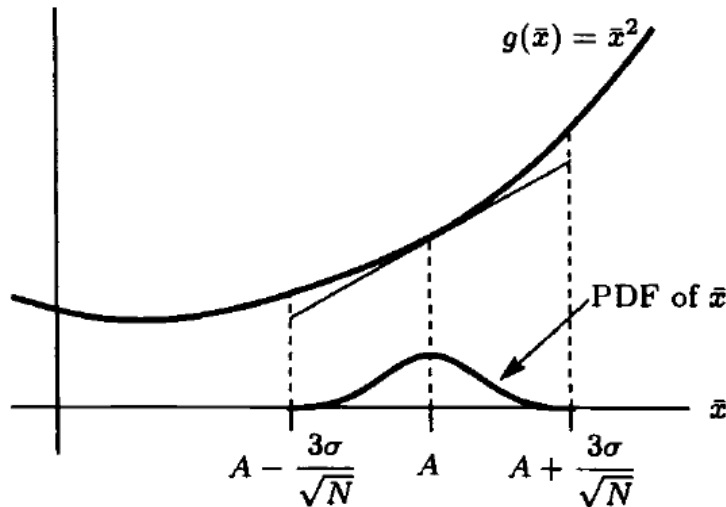
$$\bar{x} = \frac{1}{N} \sum_{n=0}^{N-1} x[n], \quad \bar{x} \sim \mathcal{N}\left(A, \frac{\sigma^2}{N}\right)$$

$\bar{x}^2$  is approximately unbiased

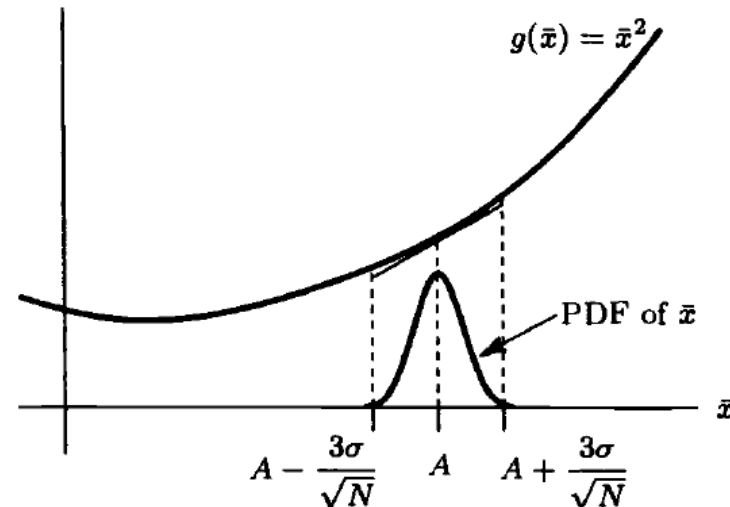
CRLB for  $A^2$ :

$$\text{var}(\widehat{A^2}) \geq \frac{(2A)^2}{N/\sigma^2} = \frac{4A^2\sigma^2}{N}$$

$$\begin{aligned} \text{var}(\bar{x}^2) &= E(\bar{x}^4) - E^2(\bar{x}^2) \\ &= \frac{4A^2\sigma^2}{N} + \frac{2\sigma^4}{N^2} \quad \text{as } N \rightarrow \infty, \text{var}(\bar{x}^2) \rightarrow \frac{4A^2\sigma^2}{N}. \end{aligned}$$



(a) Small  $N$



(b) Large  $N$

$$\bar{x} = \frac{1}{N} \sum_{n=0}^{N-1} x[n], \quad \bar{x} \sim \mathcal{N}\left(A, \frac{\sigma^2}{N}\right)$$

CRLB for  $A^2$ :

$$\text{var}(\widehat{A^2}) \geq \frac{(2A)^2}{N/\sigma^2} = \frac{4A^2\sigma^2}{N}$$

$$g(\bar{x}) = \bar{x}^2$$

Linearize  $g$  about  $A$ :

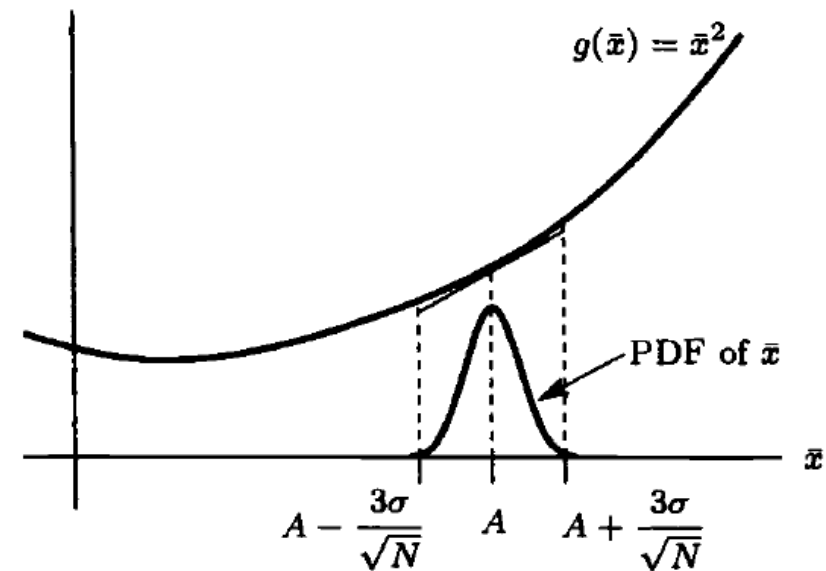
$$g(\bar{x}) \approx g(A) + \frac{dg(A)}{dA}(\bar{x} - A)$$

Under this approximation:

$$E[g(\bar{x})] = g(A) = A^2$$

$$\begin{aligned} \text{var}[g(\bar{x})] &= \left[ \frac{dg(A)}{dA} \right]^2 \text{var}(\bar{x}) \\ &= \frac{(2A)^2 \sigma^2}{N} \\ &= \frac{4A^2 \sigma^2}{N} \end{aligned}$$

**Asymptotically Efficient**



# **CRLB for Vector Parameters**

Vector parameter

$\hat{\theta}$ , an unbiased estimate of  $\theta$ .

$$\theta = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_p \end{bmatrix}$$

CRLB:

$$\text{var}(\hat{\theta}_i) \geq [\mathbf{I}^{-1}(\boldsymbol{\theta})]_{ii}$$

$I(\theta)$ , Fisher information matrix

$$[\mathbf{I}(\boldsymbol{\theta})]_{ij} = -E \left[ \frac{\partial^2 \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \theta_i \partial \theta_j} \right]$$

$$x[n] = A + w[n], \quad \text{where } w[n] \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$$

Unknowns:  $A$  and  $\sigma^2$

$$\ln p(\mathbf{x}; \boldsymbol{\theta}) = -\frac{N}{2} \ln 2\pi - \frac{N}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2$$

$$\mathbf{I}(\boldsymbol{\theta}) = \begin{bmatrix} -E \left[ \frac{\partial^2 \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial A^2} \right] & -E \left[ \frac{\partial^2 \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial A \partial \sigma^2} \right] \\ -E \left[ \frac{\partial^2 \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \sigma^2 \partial A} \right] & -E \left[ \frac{\partial^2 \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \sigma^2 \partial \sigma^2} \right] \end{bmatrix}$$

$$= \begin{bmatrix} \frac{N}{\sigma^2} & 0 \\ 0 & \frac{N}{2\sigma^4} \end{bmatrix}$$

$$\begin{aligned} \text{var}(\hat{A}) &\geq \frac{\sigma^2}{N} \\ \text{var}(\hat{\sigma}^2) &\geq \frac{2\sigma^4}{N} \end{aligned}$$

$$\frac{\partial \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial A} = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)$$

$$\frac{\partial \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \sigma^2} = -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{n=0}^{N-1} (x[n] - A)^2$$

$$\frac{\partial^2 \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial A^2} = -\frac{N}{\sigma^2}$$

$$\frac{\partial^2 \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial A \partial \sigma^2} = -\frac{1}{\sigma^4} \sum_{n=0}^{N-1} (x[n] - A)$$

$$\frac{\partial^2 \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \sigma^2 \partial \sigma^2} = \frac{N}{2\sigma^4} - \frac{1}{\sigma^6} \sum_{n=0}^{N-1} (x[n] - A)^2.$$

$$x[n] = A + Bn + w[n] \quad n = 0, 1, \dots, N-1$$

$$\boldsymbol{\theta} = [A \ B]^T$$

$$p(\mathbf{x}; \boldsymbol{\theta}) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A - Bn)^2 \right\}$$

$$\frac{\partial \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial A} = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} (x[n] - A - Bn)$$

$$\frac{\partial \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial B} = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} (x[n] - A - Bn)n$$

$$\frac{\partial^2 \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial A^2} = -\frac{N}{\sigma^2}$$

$$\frac{\partial^2 \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial A \partial B} = -\frac{1}{\sigma^2} \sum_{n=0}^{N-1} n$$

$$\frac{\partial^2 \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial B^2} = -\frac{1}{\sigma^2} \sum_{n=0}^{N-1} n^2$$

$$\begin{aligned} \mathbf{I}(\boldsymbol{\theta}) &= \frac{1}{\sigma^2} \begin{bmatrix} N & \sum_{n=0}^{N-1} n \\ \sum_{n=0}^{N-1} n & \sum_{n=0}^{N-1} n^2 \end{bmatrix} \\ &= \frac{1}{\sigma^2} \begin{bmatrix} N & \frac{N(N-1)}{2} \\ \frac{N(N-1)}{2} & \frac{N(N-1)(2N-1)}{6} \end{bmatrix} \end{aligned}$$

$$\mathbf{I}^{-1}(\boldsymbol{\theta}) = \sigma^2 \begin{bmatrix} \frac{2(2N-1)}{N(N+1)} & -\frac{6}{N(N+1)} \\ -\frac{6}{N(N+1)} & \frac{12}{N(N^2-1)} \end{bmatrix}$$

$$\text{var}(\hat{A}) \geq \frac{2(2N-1)\sigma^2}{N(N+1)}$$

$$\text{var}(\hat{B}) \geq \frac{12\sigma^2}{N(N^2-1)}$$

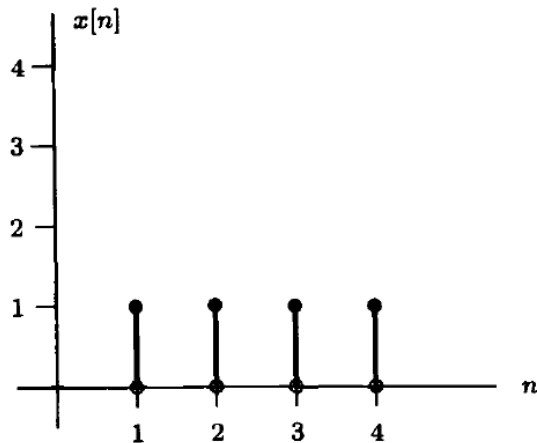
## 1. CRLB always increases as we estimate more parameters

- When only  $A$  is unknown,  $\text{var}(\hat{A}) \geq \frac{\sigma^2}{N}$ .
- When both  $A$  and  $B$  are unknown,  $\text{var}(\hat{A}) \geq \frac{2(2N-1)\sigma^2}{N(N+1)}$ .
- For  $N \geq 2$ ,  $\frac{2(2N-1)\sigma^2}{N(N+1)} > \frac{\sigma^2}{N}$ .

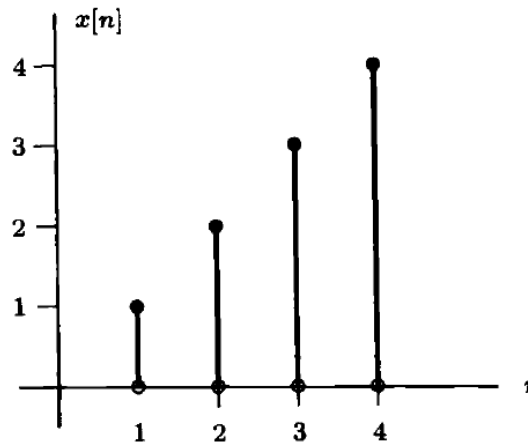
## 2. Some parameters are more sensitive than others

$$\frac{\text{CRLB}(\hat{A})}{\text{CRLB}(\hat{B})} = \frac{(2N-1)(N-1)}{6} > 1 \quad \text{for } N \geq 3. \quad \text{i.e., B is easier to estimate}$$

$$x[n] = A + Bn + w[n] \quad n = 0, 1, \dots, N-1$$



(a)  $A = 0, B = 0$  to  $A = 1, B = 0$



(b)  $A = 0, B = 0$  to  $A = 0, B = 1$

$$\Delta x[n] \approx \frac{\partial x[n]}{\partial A} \Delta A = \Delta A$$

$$\Delta x[n] \approx \frac{\partial x[n]}{\partial B} \Delta B = n \Delta B$$

**Theorem 3.1 (Cramer-Rao Lower Bound - Scalar Parameter)** *It is assumed that the PDF  $p(\mathbf{x}; \theta)$  satisfies the “regularity” condition*

$$E \left[ \frac{\partial \ln p(\mathbf{x}; \theta)}{\partial \theta} \right] = 0 \quad \text{for all } \theta$$

*where the expectation is taken with respect to  $p(\mathbf{x}; \theta)$ . Then, the variance of any unbiased estimator  $\hat{\theta}$  must satisfy*

$$\text{var}(\hat{\theta}) \geq \frac{1}{-E \left[ \frac{\partial^2 \ln p(\mathbf{x}; \theta)}{\partial \theta^2} \right]} \quad (3.6)$$

*where the derivative is evaluated at the true value of  $\theta$  and the expectation is taken with respect to  $p(\mathbf{x}; \theta)$ . Furthermore, an unbiased estimator may be found that attains the bound for all  $\theta$  if and only if*

$$\frac{\partial \ln p(\mathbf{x}; \theta)}{\partial \theta} = I(\theta)(g(\mathbf{x}) - \theta) \quad (3.7)$$

*for some functions  $g$  and  $I$ . That estimator, which is the MVU estimator, is  $\hat{\theta} = g(\mathbf{x})$ , and the minimum variance is  $1/I(\theta)$ .*



**Theorem 3.2 (Cramer-Rao Lower Bound - Vector Parameter)** *It is assumed that the PDF  $p(\mathbf{x}; \boldsymbol{\theta})$  satisfies the “regularity” conditions*

$$E \left[ \frac{\partial \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right] = \mathbf{0} \quad \text{for all } \boldsymbol{\theta}$$

*where the expectation is taken with respect to  $p(\mathbf{x}; \boldsymbol{\theta})$ . Then, the covariance matrix of any unbiased estimator  $\hat{\boldsymbol{\theta}}$  satisfies*

$$\mathbf{C}_{\hat{\boldsymbol{\theta}}} - \mathbf{I}^{-1}(\boldsymbol{\theta}) \geq \mathbf{0} \quad (3.24)$$

*where  $\geq \mathbf{0}$  is interpreted as meaning that the matrix is positive semidefinite. The Fisher information matrix  $\mathbf{I}(\boldsymbol{\theta})$  is given as*

$$[\mathbf{I}(\boldsymbol{\theta})]_{ij} = -E \left[ \frac{\partial^2 \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \theta_i \partial \theta_j} \right]$$

*where the derivatives are evaluated at the true value of  $\boldsymbol{\theta}$  and the expectation is taken with respect to  $p(\mathbf{x}; \boldsymbol{\theta})$ . Furthermore, an unbiased estimator may be found that attains the bound in that  $\mathbf{C}_{\hat{\boldsymbol{\theta}}} = \mathbf{I}^{-1}(\boldsymbol{\theta})$  if and only if*

$$\frac{\partial \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \mathbf{I}(\boldsymbol{\theta})(\mathbf{g}(\mathbf{x}) - \boldsymbol{\theta}) \quad (3.25)$$

*for some  $p$ -dimensional function  $\mathbf{g}$  and some  $p \times p$  matrix  $\mathbf{I}$ . That estimator, which is the MVU estimator, is  $\hat{\boldsymbol{\theta}} = \mathbf{g}(\mathbf{x})$ , and its covariance matrix is  $\mathbf{I}^{-1}(\boldsymbol{\theta})$ .*

## Vector Parameter CRLB for Transformations

$\boldsymbol{\alpha} = \mathbf{g}(\boldsymbol{\theta})$   $r$ -dimensional function

Jacobian matrix ( $r \times p$ ):

$$\mathbf{C}_{\hat{\alpha}} - \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \mathbf{I}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{g}(\boldsymbol{\theta})^T}{\partial \boldsymbol{\theta}} \geq \mathbf{0}$$

$$\frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \begin{bmatrix} \frac{\partial g_1(\boldsymbol{\theta})}{\partial \theta_1} & \frac{\partial g_1(\boldsymbol{\theta})}{\partial \theta_2} & \cdots & \frac{\partial g_1(\boldsymbol{\theta})}{\partial \theta_p} \\ \frac{\partial g_2(\boldsymbol{\theta})}{\partial \theta_1} & \frac{\partial g_2(\boldsymbol{\theta})}{\partial \theta_2} & \cdots & \frac{\partial g_2(\boldsymbol{\theta})}{\partial \theta_p} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial g_r(\boldsymbol{\theta})}{\partial \theta_1} & \frac{\partial g_r(\boldsymbol{\theta})}{\partial \theta_2} & \cdots & \frac{\partial g_r(\boldsymbol{\theta})}{\partial \theta_p} \end{bmatrix}$$

$x[n] = A + w[n]$ , where  $w[n] \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$ .

- Unknowns:  $A$  and  $\sigma^2$
- Estimate  $\alpha = \frac{A^2}{\sigma^2}$  (SNR)

$$\begin{aligned} \frac{\partial g(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} &= \begin{bmatrix} \frac{\partial g(\boldsymbol{\theta})}{\partial \theta_1} & \frac{\partial g(\boldsymbol{\theta})}{\partial \theta_2} \end{bmatrix} = \begin{bmatrix} \frac{\partial g(\boldsymbol{\theta})}{\partial A} & \frac{\partial g(\boldsymbol{\theta})}{\partial \sigma^2} \end{bmatrix} \\ &= \begin{bmatrix} \frac{2A}{\sigma^2} & -\frac{A^2}{\sigma^4} \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \mathbf{I}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{g}(\boldsymbol{\theta})^T}{\partial \boldsymbol{\theta}} &= \begin{bmatrix} \frac{2A}{\sigma^2} & -\frac{A^2}{\sigma^4} \end{bmatrix} \begin{bmatrix} \frac{\sigma^2}{N} & 0 \\ 0 & \frac{2\sigma^4}{N} \end{bmatrix} \begin{bmatrix} \frac{2A}{\sigma^2} \\ -\frac{A^2}{\sigma^4} \end{bmatrix} \\ &= \frac{4A^2}{N\sigma^2} + \frac{2A^4}{N\sigma^4} \\ &= \frac{4\alpha + 2\alpha^2}{N}. \end{aligned}$$

$$\text{var}(\hat{\alpha}) \geq \frac{4\alpha + 2\alpha^2}{N}$$

## CRLB for the General Gaussian Case

$$\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}(\boldsymbol{\theta}), \mathbf{C}(\boldsymbol{\theta}))$$

$$\begin{aligned} [\mathbf{I}(\boldsymbol{\theta})]_{ij} &= \left[ \frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})}{\partial \theta_i} \right]^T \mathbf{C}^{-1}(\boldsymbol{\theta}) \left[ \frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})}{\partial \theta_j} \right] \\ &+ \frac{1}{2} \text{tr} \left[ \mathbf{C}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_i} \mathbf{C}^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_j} \right] \end{aligned}$$

Appendix 3C

$$\frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})}{\partial \theta_i} = \begin{bmatrix} \frac{\partial [\boldsymbol{\mu}(\boldsymbol{\theta})]_1}{\partial \theta_i} \\ \frac{\partial [\boldsymbol{\mu}(\boldsymbol{\theta})]_2}{\partial \theta_i} \\ \vdots \\ \frac{\partial [\boldsymbol{\mu}(\boldsymbol{\theta})]_N}{\partial \theta_i} \end{bmatrix} \quad \frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_i} = \begin{bmatrix} \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{11}}{\partial \theta_i} & \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{12}}{\partial \theta_i} & \cdots & \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{1N}}{\partial \theta_i} \\ \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{21}}{\partial \theta_i} & \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{22}}{\partial \theta_i} & \cdots & \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{2N}}{\partial \theta_i} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{N1}}{\partial \theta_i} & \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{N2}}{\partial \theta_i} & \cdots & \frac{\partial [\mathbf{C}(\boldsymbol{\theta})]_{NN}}{\partial \theta_i} \end{bmatrix}$$

$$\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}(\theta), \mathbf{C}(\theta)) \quad (\text{scalar parameter})$$

$$\begin{aligned} I(\theta) &= \left[ \frac{\partial \boldsymbol{\mu}(\theta)}{\partial \theta} \right]^T \mathbf{C}^{-1}(\theta) \left[ \frac{\partial \boldsymbol{\mu}(\theta)}{\partial \theta} \right] \\ &+ \frac{1}{2} \text{tr} \left[ \left( \mathbf{C}^{-1}(\theta) \frac{\partial \mathbf{C}(\theta)}{\partial \theta} \right)^2 \right] \end{aligned}$$

$x[n] = A + w[n]$ , where

- $w[n] \sim \mathcal{N}(0, \sigma^2)$
- $A \sim \mathcal{N}(0, \sigma_A^2)$

$$\begin{aligned} [\mathbf{C}(\sigma_A^2)]_{ij} &= E[x[i-1]x[j-1]] \\ &= E[(A + w[i-1])(A + w[j-1])] \\ &= \sigma_A^2 + \sigma^2 \delta_{ij}. \end{aligned}$$

$$\mathbf{C}(\sigma_A^2) = \sigma_A^2 \mathbf{1}\mathbf{1}^T + \sigma^2 \mathbf{I}$$

Matrix inversion lemma:

$$(\mathbf{A} + \mathbf{BCD})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{B}(\mathbf{DA}^{-1}\mathbf{B} + \mathbf{C}^{-1})^{-1}\mathbf{DA}^{-1}$$

$$\mathbf{C}^{-1}(\sigma_A^2) = \frac{1}{\sigma^2} \left( \mathbf{I} - \frac{\sigma_A^2}{\sigma^2 + N\sigma_A^2} \mathbf{1}\mathbf{1}^T \right)$$

Woodbury identity:

$$(\mathbf{A} + \mathbf{uu}^T)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{uu}^T\mathbf{A}^{-1}}{1 + \mathbf{u}^T\mathbf{A}^{-1}\mathbf{u}}$$

$$\frac{\partial \mathbf{C}(\sigma_A^2)}{\partial \sigma_A^2} = \mathbf{1}\mathbf{1}^T$$

$$\mathbf{C}^{-1}(\sigma_A^2) \frac{\partial \mathbf{C}(\sigma_A^2)}{\partial \sigma_A^2} = \frac{1}{\sigma^2 + N\sigma_A^2} \mathbf{1}\mathbf{1}^T$$

$$\begin{aligned} I(\theta) &= \left[ \frac{\partial \boldsymbol{\mu}(\theta)}{\partial \theta} \right]^T \mathbf{C}^{-1}(\theta) \left[ \frac{\partial \boldsymbol{\mu}(\theta)}{\partial \theta} \right] \\ &\quad + \frac{1}{2} \text{tr} \left[ \left( \mathbf{C}^{-1}(\theta) \frac{\partial \mathbf{C}(\theta)}{\partial \theta} \right)^2 \right] \end{aligned}$$

$$\begin{aligned} I(\sigma_A^2) &= \frac{1}{2} \text{tr} \left[ \left( \frac{1}{\sigma^2 + N\sigma_A^2} \right)^2 \mathbf{1}\mathbf{1}^T \mathbf{1}\mathbf{1}^T \right] \\ &= \frac{N}{2} \left( \frac{1}{\sigma^2 + N\sigma_A^2} \right)^2 \text{tr}(\mathbf{1}\mathbf{1}^T) \\ &= \frac{1}{2} \left( \frac{N}{\sigma^2 + N\sigma_A^2} \right)^2 \end{aligned}$$

$$\text{var}(\sigma_A^2) \geq 2 \left( \sigma_A^2 + \frac{\sigma^2}{N} \right)^2$$